

Curriculum Vitae

Koustubh Bhattacharjee

Former US High School Physics Teacher| Remote Physics and Math Tutor(UK,US)| Former Research Assistant at UT Dallas
Washington, D.C., USA(Indian Citizen)

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Objective

Passionate physics educator with experience in high school and university-level teaching, as well as research in astrophysics. With a strong academic foundation in both teaching and research, I aim to contribute to the academic excellence in India through engaging and innovative teaching methods while continuing to expand my knowledge in physics.

Education

University of Texas at Dallas, Richardson, TX, USA

MS in Physics, PhD (Discontinued)

May 2023

- Research focus on Ultra-Short Period Planets and Stellar Evolution.
- Completed advanced coursework in Electromagnetism, Classical Mechanics, and Astrophysics.

Indian Institute of Space Science and Technology (IIST), Trivandrum, India

MS in Astronomy and Astrophysics

August 2015 – May 2017

- Master's Thesis: Study of Gamma-Ray Burst Prompt Emission and Detection Scenarios.

National Institute of Technology (NIT), Silchar, India

B.Tech in Electronics and Communication Engineering

August 2011 – May 2015

Teaching Experience

Multiple Agencies(India,UK)

Remote Physics and Math Tutor

Oct 2025 – present

Tutoring(1:1) students across the US and UK on Physics and Math subjects. Current/recent students have been for:

- AP physics
- GCSE math
- A levels math(pure, mechanics, statistics)
- Precalculus
- SAT math
- Algebra-II

KIPP DC, Washington, D.C., USA

High School Physics Teacher

Dec 2023 – July 2025

- Taught Physics to 11th and 12th grade students, focusing on conceptual understanding and practical applications.
- Developed curriculum and implemented phenomena-based teaching methods to engage students from diverse socio-economic backgrounds.
- Fostered a student-centered learning environment with effective classroom management techniques.

Upward Bound After School Program, Lancaster, TX, USA

Science Instructor (Part-time)

October 2023

- Taught science subjects to high school students, providing additional academic support.
- Assisted students with SAT preparation and college applications, ensuring their academic readiness.

University of Texas at Dallas, Richardson, TX

Teaching Assistant

2017 – 2020, 2022 – 2023

- Delivered lectures, graded assignments, and conducted lab sessions for undergraduate courses in Electromagnetism (E&M) and Classical Mechanics.
- Provided guidance and mentorship to students, helping them succeed in both theoretical and practical aspects of physics.

Research Experience

University of Texas at Dallas, Richardson, TX

Research Assistant

May 2019 – Dec 2021

- Investigated the origin of Ultra-Short Period Planets using the MESA stellar evolution code and Python.
- Led simulations to explore the interaction between tidal inspiral and Roche Lobe overflow in hot Jupiter systems.
- Conducted research in astrophysical modelling using parallel computing and high-performance clusters.

Indian Institute of Space Science and Technology (IIST), Trivandrum, India

Master's Thesis Researcher

2015 – 2017

- Studied the prompt emission of Gamma-Ray Bursts (GRBs) and developed detection scenarios.
- Collaborated with Dr. Reshmi Lekshmi on GRB observational data and theoretical modeling.

Skills

- **Research & Programming:** Proficient in Python (NumPy, SciPy), MESA, and Fortran90; experience with data analysis and computational simulations.
- **Astrophysical Tools:** Skilled in using IRAF, Astropy, OpenCV, and other image processing libraries for astronomical data analysis.
- **Pedagogical Skills:** Classroom management, curriculum development, and phenomena-based teaching approaches.
- **Embedded Systems:** Experience with 8051, Atmega16, and 8085 microcontrollers.
- **Web Development:** Knowledge of front-end development with JavaScript, React, and WebGL (Three.js).
- **Cloud Development:** Basic proficiency in cloud-based learning platforms and development environments.

Projects

- **Doctoral Thesis (2019 – Present):** Simulation of Ultra-Short Period Planets using MESA stellar evolution code and Python.
- **Master's Thesis (2016 – 2017):** Study of On-Axis and Off-Axis Characteristics of Prompt GRB Emission.
- **Undergraduate Projects:**

- Railway Accident Prevention System using ATMEGA32.
 - Water Level Detection and Alarm System using Atmega16.
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Achievements

- **JEST (Joint Entrance Screening Test):**
 - Achieved a ~90 percentile score, securing an All India rank of ~270 (2015) and ~320 (2017), qualifying for prestigious research programs in astrophysics and physics.
 - **Doctoral Proposal Defense:** Successfully defended my doctoral research proposal at UT Dallas in January 2022.
 - **Master's Thesis Defense:** Successfully defended my Master's thesis on Gamma-Ray Bursts in 2017.
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Internships

- **Hewlett-Packard (HP), 2013:** Provided end-to-end solutions for enterprise network infrastructure.
- **CETPA Infotech, Noida, 2012:** Developed a gesture-controlled robot using the 8051 microcontroller and accelerometer.